

RESEARCH PROJECT DEFINITION

Geometry-Aware POD-Galerkin PINN ROM for Inverse Stenosis Analysis

Working research title: Physics-Informed Inverse Reduced-Order Modeling of Stenotic Hemodynamics

Independent Research Project

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Project definition and planning document. Not a clinical protocol, diagnostic system, or claim of validated patient-specific performance.

Document Control

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0.1	10 Jul 2026	Initial definition	Initial research thesis, scope, methodology, data strategy, validation framework, risks, and next actions.	E. Gregorius

Document Purpose and Use

This document establishes the initial definition of an independent research and software project. It is intended to organize the work, preserve the original research rationale, define what will and will not be claimed, and provide an accessible technical overview for engineers, researchers, and prospective faculty reviewers. It is not a substitute for the engineering notebook. The notebook will contain the dated record of actual experiments, design changes, source evaluation, code decisions, and unsuccessful attempts. A later technical report or manuscript, if warranted by results, will present the final methods and evidence.

The research direction has been selected, but feasibility and accuracy have not yet been demonstrated. Statements in this document therefore distinguish between the proposed method, the initial evidence supporting feasibility, and results that must still be generated. The document may be revised when data access, geometric representation, reduced-basis behavior, or inverse identifiability is better understood.

Independent research status.

The project is being initiated independently using public literature, public or appropriately licensed datasets, original code, synthetic simulations, and personal computing resources. It is not currently an approved thesis, sponsored research topic, clinical study, or university-endorsed product. External technical guidance or future coauthorship may develop if faculty contribute substantially, but no such relationship is assumed in this version.

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Executive Summary

Arterial stenosis changes local blood velocity, pressure, wall shear stress, and downstream recirculation. Computational fluid dynamics can resolve these quantities, but repeated high-fidelity simulations become expensive when geometry, flow conditions, material assumptions, or inverse parameters must be varied. Conventional image-only neural networks can estimate anatomical narrowing from labeled scans, but they do not necessarily recover the associated physical fields or preserve the governing fluid mechanics. This project proposes a reduced-order inverse framework that places a low-dimensional representation of the Navier-Stokes equations between these two extremes.

The proposed method combines geometry-compatible proper orthogonal decomposition (POD), Galerkin projection, physics-informed neural networks (PINNs), and a constrained learned residual closure. A full-order snapshot library would first be assembled from public computational hemodynamics data, newly generated idealized simulations, or a controlled mixture of both. Geometry-varying fields would be mapped into a compatible representation before POD bases are constructed for velocity and pressure. Galerkin projection would then produce reduced equations. A neural network would estimate reduced coefficients and stenosis parameters while being penalized for violating those reduced equations. A learned closure would be introduced only after an uncorrected model is established, and would be evaluated as a correction for basis truncation, model-form discrepancy, uncertain boundary conditions, and geometric effects not represented in the initial reduced system.

The initial controlled parameter space will emphasize stenosis severity, nondimensional stenosis length, Reynolds number, and blood rheology. Stenosis location, eccentricity, curvature, pulsatile inlet conditions, and additional measurement modalities are later extensions. The primary reconstructed fields will be velocity and pressure. Wall shear stress, pressure drop, recirculation measures, and oscillatory shear index will be treated as derived quantities whenever the required spatial or temporal resolution is available.

The central research question is whether a geometry-aware POD-Galerkin PINN can operate as a reusable inverse surrogate that reconstructs stenosis parameters and hemodynamic fields from limited observations at substantially lower online cost than full-order CFD, while retaining accuracy and reliability competitive with the available high-fidelity data. The study will also determine which measurement types make stenosis severity identifiable, whether reduced physics lowers the amount of training data required, how geometry variation affects the reduced basis, and whether a constrained residual closure adds value beyond a data-only POD model.

The intended outcome is a documented method, verified software pipeline, and demonstration interface rather than a clinical product. A negative result remains useful if the experiments show that geometric variability prevents a compact global basis, that sparse observations do not identify lesion parameters, or that the physics-informed formulation does not outperform simpler baselines. The project will preserve those results rather than retroactively changing the hypothesis to fit the outcome.

1. Project Thesis and Research Position

1.1 Project thesis

The project will investigate the following thesis:

Research hypothesis.

A geometry-aware POD-Galerkin PINN with a constrained learned residual closure can recover stenosis parameters and full velocity and pressure fields from sparse, noisy, or incomplete observations more accurately and reliably than purely data-driven reduced models, while reducing the online computational cost relative to full-order CFD.

The hypothesis contains two distinct claims that must be tested separately. The first is computational: a trained reduced model should evaluate new cases much faster than a full-order solver. The second is inferential: reduced physical constraints should improve the recovery of unknown lesion parameters or fields when

observations are incomplete. A speedup without acceptable field error is not sufficient. Likewise, an accurate inverse result that requires patient-by-patient retraining or a computational cost similar to CFD would not satisfy the intended use.

1.2 Primary research question

Can a geometry-aware POD-Galerkin PINN be trained as a reusable inverse surrogate that reconstructs stenosis parameters and velocity and pressure fields from limited hemodynamic observations at substantially lower online cost than full-order CFD, while remaining accurate within a defined family of arterial geometries and flow conditions?

1.3 Secondary research questions

- Which observation types and sensor locations make stenosis severity, length, or location identifiable?
- How much online speedup is achieved relative to the selected full-order reference, including pre-processing and reconstruction costs?
- Does inclusion of reduced Galerkin physics reduce the amount of training data required relative to POD interpolation and data-only POD neural networks?
- How strongly do geometry registration, POD dimension, and local flow separation control reconstruction error?
- Which measured or derived hemodynamic quantities are most informative for inverse severity estimation, and which correlations are stable across geometry and flow variations?
- Does a constrained residual closure improve performance on unresolved physics or unseen in-range geometries without replacing the reduced equations?
- Can missing modalities be represented explicitly so that uncertainty increases when the available evidence is insufficient?

1.4 Intended contribution

The principal contribution is not the invention of POD, Galerkin projection, PINNs, or computational hemodynamics. Each is established independently. The contribution sought here is an integrated, testable implementation for inverse stenosis analysis in which geometry-varying full-order data are reduced to compatible coordinates, governing equations constrain the learned surrogate, incomplete observations are handled explicitly, and a residual correction is evaluated against uncorrected and data-only alternatives. The work is intended to show where the combination is effective and where it fails.

- A geometry-compatible representation of stenotic-flow snapshots across a controlled range of lesion shapes.
- An inverse reduced-order model that estimates lesion parameters and reconstructs velocity and pressure.
- A controlled comparison of data-only and Galerkin-informed reduced models under sparse, noisy, and missing observations.
- A constrained residual or discrepancy model introduced after the baseline reduced physics has been verified.
- Uncertainty or reliability outputs that distinguish confident interpolation from ambiguous or out-of-domain cases.
- A reproducible codebase and demonstration interface after the method has been validated.

1.5 Research posture

The project is framed as an investigation rather than a predetermined demonstration of superiority. The proposed architecture may prove unnecessary if a simpler POD surrogate performs equally well. It may also prove unsuitable if the flow manifold is not sufficiently low dimensional across the chosen geometry family. Those outcomes would narrow the contribution, but they would not invalidate the work if the comparison is performed rigorously and documented completely.

2. Background and Motivation

2.1 Computational hemodynamics of stenosis

Computational hemodynamics applies numerical fluid mechanics to blood flow in vessels, cardiac chambers, and blood-contacting devices. Velocity and pressure are the primitive flow variables for incompressible Navier-Stokes models. Clinically or biologically relevant quantities such as pressure loss, wall shear stress (WSS), oscillatory shear index (OSI), and recirculation are calculated from those fields. Image-based models can also incorporate realistic anatomy, moving boundaries, non-Newtonian viscosity, and uncertain inlet or outlet conditions, but each added feature increases the data and computational requirements [3].

Zhou et al. provide a useful initial stenosis benchmark: an axisymmetric tube with a cosine-shaped narrowing, fixed Reynolds number, and both Newtonian and Carreau blood models [2]. Their severity parameter is based on area reduction,

$$\eta = 1 - (R_t/R)^2, \quad (1)$$

where R is the unobstructed radius and R_t is the throat radius. In that controlled problem, increasing severity produces a larger post-stenotic separation region, a larger pressure drop, and a marked increase in peak WSS. The benchmark is deliberately simplified, but it provides a reproducible starting point for checking geometry generation, rheology, mesh convergence, and post-processing before more complex data are introduced.

The Reynolds number will be treated as an explicit parameter because it changes the balance between inertia and viscosity:

$$Re = \rho U D / \mu, \quad (2)$$

with density ρ , characteristic velocity U , diameter D , and dynamic viscosity μ . When a shear-dependent rheology is used, μ becomes a function of local shear rate. The initial project will not assume that one rheology model is always required. Instead, Newtonian and Carreau-type models will be considered as separate levels of fidelity and as possible sources of discrepancy for the closure study.

2.2 Why reduced-order modeling is relevant

Many-query applications motivate reduced-order modeling. Examples include parameter sweeps, uncertainty propagation, optimization, repeated inverse estimation, and interactive exploration. In these settings, the cost of generating a full-order training library can be justified if the trained model is evaluated many times afterward. The reduced model does not eliminate the full-order cost; it moves that cost into an offline stage and seeks inexpensive online evaluation.

POD represents a field using a small number of spatial modes. For velocity and pressure, a centered approximation can be written as

$$u(x,t;\mu) \approx \bar{u}(x) + \sum_{i=1}^{N_u} a_i(t;\mu) \varphi_i(x), \quad (3a)$$

$$p(x,t;\mu) \approx \bar{p}(x) + \sum_{j=1}^{N_p} b_j(t;\mu) \chi_j(x). \quad (3b)$$

The coefficients a and b replace the full mesh fields during online evaluation. POD alone is a compression method. It becomes a projection-based reduced model when the governing equations are projected onto the reduced spaces. Non-intrusive alternatives instead learn the coefficients from parameters or geometry. Hesthaven and Ubbiali showed that neural networks can approximate parameter-to-reduced-coefficient maps for nonlinear problems [5].

2.3 POD-Galerkin PINNs for inverse problems

Hijazi et al. combine POD-Galerkin reduction with PINNs for inverse Navier-Stokes problems [1]. Their method assumes simulated field snapshots, constructs POD bases and reduced operators, and trains neural networks whose outputs are reduced flow variables. The training objective includes both reduced data and

residuals of the Galerkin equations. Unknown quantities such as viscosity or inlet velocity can be introduced as trainable parameters. After training, the model can perform rapid forward evaluations without retraining for every parameter value inside the learned domain.

The proposed stenosis project follows that structure before introducing new elements. The first implementation should reproduce the logic of the published method as closely as practical: field projection, reduced residual construction, data and physics losses, and inverse parameter estimation. Geometry-aware snapshot mapping, lesion-parameter inversion, missing observations, and the learned residual closure should be added only after the base implementation is functioning.

2.4 Geometry is the central extension

Stenosis severity changes the computational domain, not only a scalar coefficient in a fixed equation. Standard POD expects snapshots with compatible degrees of freedom. Directly stacking fields from unrelated meshes can mix discretization differences with physical variability. Ye et al. addressed this problem by registering similar vascular surfaces, mapping hemodynamic fields through diffeomorphisms to a reference domain, constructing geometry-informed snapshots, and then applying a non-intrusive POD-RBF model to stenosis and bifurcation cases [6]. Their work establishes that geometry-compatible reduced representations are feasible, while also showing that geometry handling is a substantive part of the model rather than a minor preprocessing step.

This project will not assume in advance that one global POD basis is sufficient. If singular-value decay or reconstruction tests show that mild and severe stenoses occupy different local manifolds, local or clustered bases may be introduced. A reference-domain mapping, common mesh topology, operator interpolation, or another compatible representation will be selected based on the first data audit and benchmark experiments.

2.5 Inverse hemodynamics from limited measurements

PINNs have already been used to estimate cardiovascular parameters and recover velocity or pressure from limited measurements. Kissas et al. inferred arterial pressure from non-invasive 4D-flow MRI data using physics-informed learning [8]. Arzani et al. reconstructed near-wall blood flow from sparse observations [9]. Garay et al. estimated reduced blood-flow parameters and full velocity fields from scattered noisy measurements in steady and transient aortic examples [7]. These studies support the feasibility of physics-guided inverse reconstruction, but they also show that performance depends on measurement quality, flow complexity, and parameter count.

More recent work has used PINNs to correct artifacts in 4D-flow MRI of an in-vitro stenosis and to assimilate mean pressure and Reynolds stresses [10]. That application is close to the proposed correction concept, but it operates as a full-field medical-flow assimilation method rather than a geometry-aware POD-Galerkin inverse surrogate. The overlap is therefore informative rather than disqualifying: it identifies an active research area and a relevant benchmark for future comparison.

2.6 Relationship to existing data-driven hemodynamics

Data-driven surrogates have already shown that patient-derived geometry can be mapped to hemodynamic quantities much faster than full CFD. Yevtushenko et al. trained a neural network on patient-specific aortic-valve CFD cases and predicted centerline pressure and WSS with reported median relative differences of 6.0% and 4.9%, respectively, on an unseen test subset [11]. That study is important for this project because it demonstrates both the value of an offline-to-online surrogate and the strength of a well-designed data-only baseline. The present work must therefore show a specific benefit from reduced physics, inverse capability, geometry compatibility, missing-data handling, or uncertainty—not merely fast neural prediction.

3. Research Problem Definition

3.1 Forward and inverse tasks

The project contains a forward task and an inverse task. The forward task maps geometry and flow parameters to reduced coefficients and reconstructed fields. The inverse task maps available observations to a distribution over lesion parameters and, when supported by the evidence, complete velocity and pressure fields.

$$\text{Forward: } (\mu, t) \rightarrow (a, b) \rightarrow (u, p), \quad (4)$$

$$\text{Inverse: } (y_{\text{obs}}, m) \rightarrow p(\theta_{\text{stenosis}}, a, b \mid y_{\text{obs}}, m), \quad (5)$$

Here μ denotes known physical and geometric parameters, θ_{stenosis} denotes unknown lesion parameters, y_{obs} denotes measurements, and m is a mask indicating which observations are present. The inverse formulation should not treat an unmeasured quantity as zero. Missingness must be encoded explicitly, and uncertainty should increase when several lesion configurations are consistent with the available data.

3.2 Initial controlled parameter domain

Parameter	Symbol	Initial role	Notes
Stenosis severity	η	Primary geometric parameter and first inverse target	Area-based severity for the initial axisymmetric or registered geometry family.
Nondimensional stenosis length	L_s/D	Secondary geometry parameter	Included when data coverage is adequate; may initially be fixed during the first severity-only experiment.
Reynolds number	Re	Flow-condition parameter	Represents changes in characteristic flow rate, diameter, and viscosity.
Blood rheology	r	Model choice or parameter set	Newtonian baseline; Carreau-type rheology considered when data and reduced-operator treatment permit.

Table 1. Initial parameter domain. Ranges will be established after the data and benchmark audit.

3.3 Later geometric and physiological parameters

Later extensions may include lesion location, eccentricity, vessel curvature, diameter variation, pulsatile waveform descriptors, and uncertain inlet or outlet conditions. These parameters are not promised deliverables in the initial version. They will be added only when the first geometry-aware basis and inverse severity model are stable. Location and eccentricity are particularly important because they change spatial correspondence and may create multiple geometries with similar global pressure signatures.

3.4 Observation model and identifiability

The model may receive pointwise or regional velocity, pressure difference, flow rate, geometry descriptors, partial field snapshots, or imaging-derived measurements. The exact first observation set will be chosen after available datasets are inspected. Systemic cuff blood pressure can provide patient context but does not uniquely determine local trans-stenotic pressure loss, WSS, or velocity. Local hemodynamics also depend on flow rate, vessel diameter, lesion shape, downstream resistance, and rheology. The inverse model must therefore be evaluated for identifiability rather than assuming that severity can always be inferred from a small set of values.

The first inverse experiments should use controlled synthetic observations drawn from known full-order cases. Sensors can then be removed, perturbed, or masked to determine which combinations retain enough information. Gaussian or modality-appropriate noise can be added at defined levels to distinguish numerical inversion from realistic measurement conditions. This use of synthetic noise does not claim to reproduce all clinical artifacts; it provides a reproducible stress test.

3.5 Primary outputs

Velocity and pressure will be treated as the primary reconstructed fields because they are the primitive variables of the incompressible flow model and the direct inputs to most derived hemodynamic metrics. The network should not be required to learn redundant outputs when they can be calculated consistently from reconstructed fields.

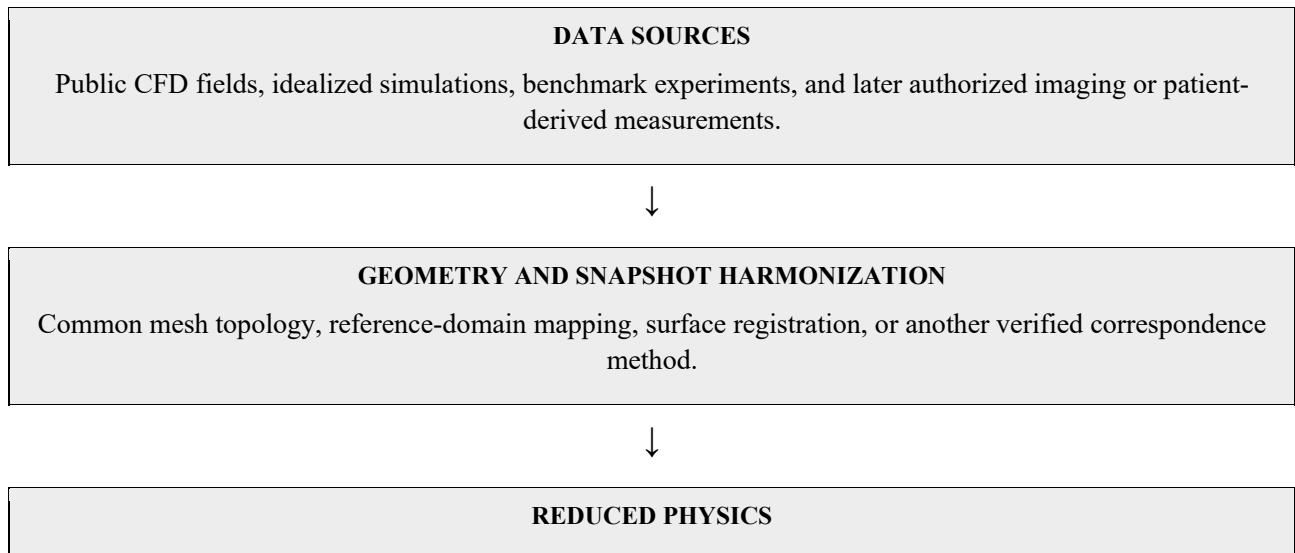
Output	Category	Treatment
Stenosis severity	Inverse parameter	Estimated directly, with uncertainty or confidence information.
Stenosis location and geometry	Later inverse parameters	Estimated when the geometry family and observations support identifiability.
Velocity field	Primary field	Reconstructed from reduced coefficients and POD basis.
Pressure field	Primary field	Reconstructed from reduced coefficients and pressure basis or compatible formulation.
Wall shear stress	Derived field	Calculated from near-wall velocity gradients; evaluated carefully because differentiation amplifies error.
Pressure drop	Derived scalar/profile	Calculated from defined upstream and downstream pressure averages.
Recirculation properties	Derived feature	Calculated from flow reversal, streamline, or separation criteria.
Oscillatory shear index	Derived temporal feature	Applicable only after a time-dependent model is established.
Prediction uncertainty	Reliability output	Method to be selected after deterministic baseline; likely ensemble or probabilistic output initially.

Table 2. Intended outputs and how each enters the initial method.

4. Proposed Methodological Framework

4.1 Overall workflow

The proposed workflow separates offline model construction from online inference. Full-order data are harmonized across geometry, compressed into POD coordinates, and used to assemble reduced operators. The inverse network then combines sparse observations with reduced physics. A residual closure is introduced as a controlled extension rather than as a substitute for the governing equations.



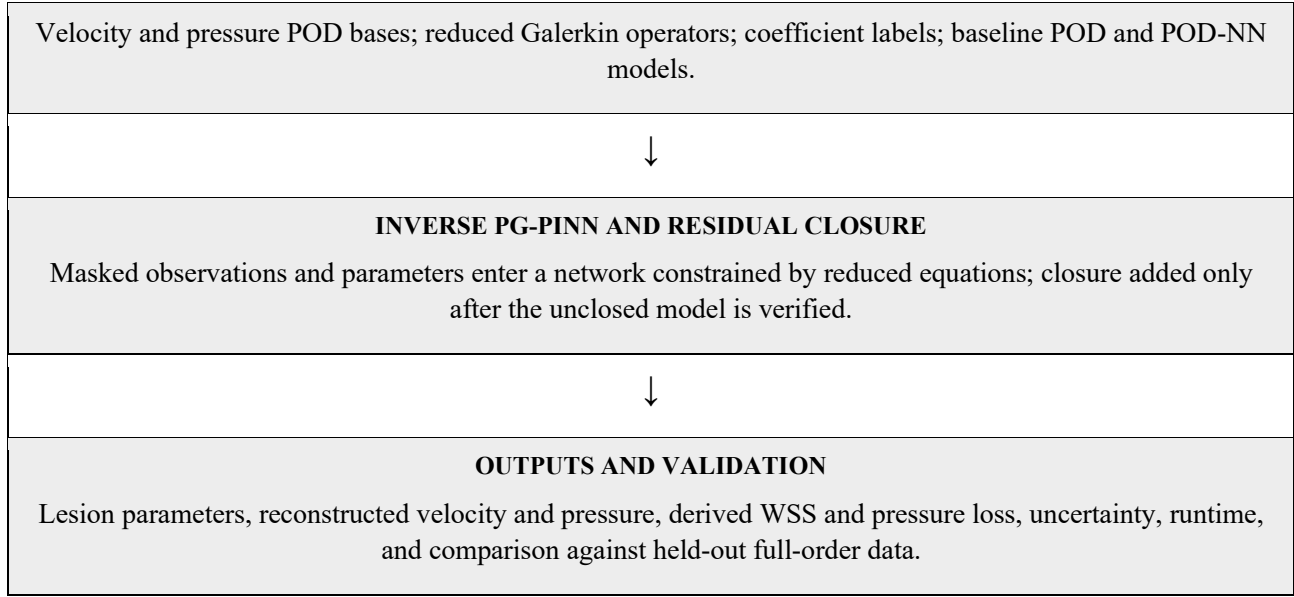


Figure 1. Proposed offline-to-online research workflow. The arrows indicate dependency rather than a fixed software architecture.

4.2 Full-order data and reference physics

The full-order reference may be assembled from three sources: public computational datasets, new idealized CFD cases, and experimental or medical-flow measurements that are legally and technically suitable. Public fields are preferred when they match the task, but the model should not be distorted to fit an unsuitable dataset. If the available public data do not include compatible velocity and pressure fields over a useful stenosis parameter range, an idealized axisymmetric or wedge simulation campaign will be generated locally and later supplemented by more complex cases.

The initial full-order model should be simple enough to verify. A Zhou-type stenosed tube provides defined geometry, laminar conditions, and Newtonian and Carreau comparisons [2]. Mesh independence, conservation, boundary-condition consistency, and comparison of pressure loss, peak WSS, and separation behavior are required before the data are used for training. A higher-fidelity extension may later use three-dimensional geometry, pulsatility, and non-Newtonian rheology, but those additions should not be made simultaneously.

4.3 Geometry-compatible snapshots

All field snapshots must share a meaningful spatial correspondence. Candidate approaches include a common topology with mesh deformation, mapping each geometry to a reference domain, or surface registration followed by volume-field transformation. The mapping must be checked independently of the neural network. Conservation error, interpolation error, wall alignment, and invertibility of the transformation should be reported. If mapping error is comparable to the final ROM error, the inverse model cannot be interpreted cleanly.

Geometry descriptors may be explicit physical parameters, compressed surface coordinates, signed-distance fields, centerline and cross-section features, or latent shape variables. The initial choice should favor interpretability and reproducibility. A geometry encoder should be introduced only when simpler parameterizations no longer cover the selected family.

4.4 POD basis construction

Separate or coupled bases will be constructed for velocity and pressure after centering and, where appropriate, nondimensionalization. The number of modes will be chosen from cumulative energy and held-out reconstruction error rather than an energy threshold alone. Pressure and WSS may require more modes than velocity because localized gradients and pressure recovery can be poorly represented by a basis optimized for global kinetic energy.

The following diagnostics will be produced before neural training: singular-value decay, cumulative retained energy, reconstruction error versus mode count, error by stenosis severity, error near the wall and throat, and evidence of clustering or regime change. If a global basis performs poorly, local POD or a mixture-of-experts formulation will be considered.

4.5 Galerkin-projected equations

Projection of the incompressible momentum and continuity equations onto the reduced bases produces a low-dimensional system. A generic representation is

$$M(\mu) \dot{a} = F_{PG}(a,b; \mu), \quad G(a,b; \mu) = 0, \quad (6)$$

where M is the reduced mass matrix, F_{PG} contains viscous, convective, pressure, boundary, and possibly rheological terms, and G represents reduced continuity or pressure stabilization. Geometry dependence may enter the operators explicitly. The first implementation will follow Hijazi et al. as closely as practical, including projected data labels and reduced residual losses [1]. Operator treatment will be revised only when the stenosis geometry or rheology requires it.

4.6 Physics-informed reduced network

The network will predict reduced coefficients and, for inverse tests, selected lesion parameters. The objective will combine data agreement, reduced physics, inverse parameter supervision where available, and regularization. A representative form is

$$L = w_{data} L_{data} + w_{phys} L_{PG} + w_{inv} L_{inv} + w_{cl} L_{cl} + w_{unc} L_{unc}. \quad (7)$$

The weights and exact terms will be justified experimentally. Physics loss should not be treated as evidence of physical validity by itself. A low residual can coexist with poor field reconstruction if the basis, boundary treatment, or operator approximation is wrong. Every learned model will therefore be compared with projected full-order coefficients and reconstructed fields.

4.7 Inverse parameter recovery

The simplest inverse task will estimate severity with all other parameters known. Complexity will then be added in a controlled order: severity with noisy measurements, severity with missing sensors, joint severity and flow-rate estimation, severity and lesion length, and finally location or eccentricity if the geometry representation supports them. This progression separates numerical training difficulty from physical non-identifiability.

Where several parameter combinations produce similar observations, the model should return a distribution or interval rather than a misleadingly precise point estimate. The project will not equate high correlation with unique physical causation. Sensor-ablation studies and posterior coverage tests will be used to identify ambiguity.

4.8 Learned residual closure

The residual closure is intended to correct a defined discrepancy after the unclosed reduced model has been evaluated. The proposed corrected system is

$$M(\mu) \dot{a} = F_{PG}(a,b; \mu) + C\psi(a,b,z,\mu), \quad (8)$$

where $C\psi$ is a learned correction and z may contain geometry or observation features. Possible targets include truncated-mode effects, unresolved non-Newtonian stresses, coarse-mesh bias, uncertain boundary conditions, mapping error, or systematic differences between idealized simulations and measured data. These sources should not be combined blindly. Initial closure experiments should use controlled paired discrepancies, such as low-versus high-mode projection or Newtonian versus Carreau reference cases, so that the learned term can be interpreted.

Candidate constraints include norm penalties, correction-to-physics ratios, zero-mean or zero-correction conditions in well-resolved cases, energy or stability penalties, and restriction of the correction to selected reduced

terms. The appropriate constraint will be researched after the baseline ROM exists. The closure will be rejected if it improves training error while degrading held-out physics, stability, or extrapolation behavior.

4.9 Missing observations and correlation structure

The model should accept an observation vector together with a binary or probabilistic mask. A simple input representation is

$$\mathbf{x}_{\text{in}} = [\mathbf{m} \odot \mathbf{x}_{\text{obs}}, \mathbf{m}], \quad (9)$$

where missing entries are masked and the mask is supplied to the model. More advanced methods, including learned imputation or latent-variable inference, may be evaluated later. The objective is not to guess missing patient data with unjustified certainty. It is to use correlations learned from the training population while preserving an uncertainty penalty when the evidence is weak or out of distribution.

4.10 Derived hemodynamic quantities

WSS, pressure drop, recirculation, and OSI will be calculated by deterministic post-processing wherever possible. WSS depends on near-wall gradients and may be sensitive to mesh mapping and basis truncation. Pressure drop depends on precisely defined measurement surfaces. OSI requires a resolved cardiac cycle. Each derived metric will therefore have its own verification test rather than being assumed correct because the global velocity error is small.

5. Data Strategy and Feasibility

5.1 Initial survey of public resources

The initial source survey indicates that useful public data exist, but they are fragmented across different tasks. A single ready-made dataset containing patient imaging, lesion labels, complete three-dimensional velocity and pressure fields, multiple flow conditions, and sparse clinical measurements has not been identified. The project should therefore be designed as a multi-source pipeline rather than assuming one dataset will satisfy every stage.

Resource	Data type	Direct value to project	Limitation	Reference
Geometry-informed stenosis/bifurcation ROM data and code	Registered vascular geometries and hemodynamic simulation data	Directly relevant to geometry-compatible POD and a reproducible implementation reference.	Published model is non-intrusive POD-RBF rather than the proposed inverse PG-PINN.	[6]
Aortic-stenosis CFD training dataset	309 CT-derived cases from 103 patients; centerline geometry, flow rate, pressure, and WSS sequences	Provides a public patient-derived geometry-to-hemodynamics dataset and data-only baseline context.	Does not provide the full volumetric velocity and pressure fields needed for a direct POD-Galerkin reproduction.	[12]
ARCADE	Annotated coronary X-ray angiography images for vessel and stenosis tasks	Useful for later imaging-derived lesion localization or geometry features.	No paired CFD fields or local pressure/velocity measurements.	[13]
FDA nozzle and blood-pump benchmark	CAD, operating conditions, experimental pressure and velocity validation data	Provides an independent V&V resource for flow solver or data-assimilation components.	Generalized medical-device geometries, not patient arterial stenosis.	[14]
Locally generated idealized stenosis cases	Full velocity, pressure, mesh, rheology, and boundary-condition control	Guarantees compatible fields and enables controlled inverse tests.	Consumes local compute and initially limits anatomical realism.	Planned

Table 3. Initial public-data assessment. Availability and licenses must be rechecked before download or redistribution.

5.2 Recommended initial data plan

1. Audit the public geometry-informed stenosis data and code from Ye et al. to determine whether full fields, meshes, parameter coverage, and licenses support direct reuse.

2. Audit the public aortic-stenosis dataset and related study to identify useful geometry descriptors, pressure/WSS targets, and a strong data-only comparison, while recognizing that it does not directly supply volumetric POD snapshots.
3. Build a small verified idealized CFD set if no public field dataset provides the required velocity-pressure pairs and geometric correspondence.
4. Use public imaging datasets only for a separate geometry or lesion-annotation branch until a legally and scientifically appropriate paired dataset is available.
5. Reserve real patient-data correction or closure training for a later phase with appropriate authorization, documentation, and a clear measurement model.

5.3 Personal-compute feasibility

Reduced-coordinate network training is feasible on a workstation-class GPU because the network outputs tens or hundreds of coefficients rather than full three-dimensional fields. Full-order CFD is more likely to be limited by CPU time, memory, mesh size, and the number of parameter cases. The initial data campaign should therefore use axisymmetric or wedge simulations, steady flow first, and automated overnight or multi-day parameter sweeps. Code architecture, data manifests, unit tests, POD analysis, and post-processing can proceed while simulations run.

A quarter-domain or symmetry model is useful only when the geometry, boundary conditions, and expected flow remain symmetric. For an axisymmetric stenosis, a two-dimensional axisymmetric formulation or thin wedge is more efficient than a quarter cylinder. Eccentric stenosis, curvature, bifurcations, and asymmetric separation require a larger domain and should be introduced only after the reduced framework is established.

5.4 Patient and medical-imaging data

Anatomical MRI, MRA, CTA, angiography, Doppler measurements, and 4D-flow MRI serve different roles. Anatomical images can define geometry and lesion location. Doppler and phase-contrast measurements provide velocity information. Pressure measurements may be systemic, local, or trans-lesional and are not interchangeable. Four-dimensional flow MRI provides time-resolved velocity fields but has finite spatial resolution, noise, and artifacts; physics-informed correction has been demonstrated in stenotic flow [10].

Real patient data may eventually train the main inverse model, the closure, or both, but only if the measurements correspond to the quantities being inferred and if data access is authorized. Patient labels alone can support classification, but they cannot replace the full velocity and pressure fields required for a Galerkin ROM. The initial project therefore excludes patient data from implementation while retaining it as a later research direction.

6. Baselines, Verification, and Validation

6.1 Required comparison models

The proposed model will not be evaluated in isolation. At minimum, the following baselines should use the same train, validation, and held-out test cases:

1. POD projection and reconstruction error, which defines the best result achievable by the selected basis for each field.
2. POD coefficient interpolation or radial-basis-function regression, providing a simple non-neural surrogate.
3. A data-only POD neural network with the same inputs and comparable capacity.
4. A conventional POD-Galerkin model or projected residual evaluation, where stable and appropriate.
5. A POD-Galerkin PINN without learned closure.
6. The proposed POD-Galerkin PINN with constrained residual closure.

A direct full-field PINN, neural operator, or image-only model may be added if it answers a specific comparison question and is computationally feasible. The project should avoid expanding the baseline list until the minimum comparison is complete.

6.2 Verification before machine learning

- Geometry parameterization reproduces the intended stenosis severity and length.
- Mesh or mapping convergence is documented for velocity, pressure drop, and WSS.
- Mass conservation and solver residuals meet defined tolerances.
- Newtonian and Carreau cases reproduce expected qualitative and quantitative trends within documented differences.
- Snapshot extraction, units, nondimensionalization, and cell ordering are reproducible.
- POD reconstruction is checked independently of the network.
- Reduced operators are tested against projected full-order time derivatives or steady residuals.

6.3 Validation metrics

Metric	Purpose	Initial interpretation
Severity MAE / absolute percentage-point error	Accuracy of η recovery	Primary inverse metric; target will be set after baseline variability is known.
Normalized lesion-location error	Accuracy of spatial inference	Used only when location is an active parameter.
Relative L2 velocity error	Global field reconstruction	Reported by case and parameter region, not only as an aggregate.
Relative L2 pressure error	Global pressure reconstruction	Supplemented by pressure-drop error because pressure offsets may be arbitrary.
Pressure-drop error	Functional flow consequence	Defined using fixed upstream/downstream surfaces.
WSS error	Near-wall derived accuracy	Peak, mean, and spatial distribution reported separately.
Inference time and speedup	Online computational value	Includes preprocessing and field reconstruction where applicable.
Noise sensitivity	Measurement robustness	Error versus specified perturbation levels.
Missing-observation sensitivity	Incomplete-data robustness	Error versus modality and sensor-ablation patterns.
Uncertainty calibration	Reliability of intervals or probabilities	Coverage and sharpness rather than confidence scores alone.

Table 4. Core evaluation metrics. Numeric acceptance thresholds are provisional until the first benchmark and baseline distributions are available.

6.4 Evaluation partitions

- Interpolation at unseen parameter combinations inside the sampled domain.
- Sparse-training experiments with progressively fewer full-order cases.
- Held-out geometry tests in which entire lesion shapes are absent from training.
- Sensor and modality ablations.
- Noise tests with repeatable random seeds and physically motivated scales.
- Out-of-domain tests used to evaluate warning behavior, not to claim generalization.

6.5 Interpretation of negative results

A result will not be considered unsuccessful solely because the proposed method fails to outperform every baseline. The study remains useful if it identifies a clear limitation: excessive POD dimension, geometry-mapping error, non-identifiable inverse parameters, unstable reduced dynamics, or a closure that overfits. The report should

preserve the original hypothesis, describe the attempted implementation, and explain why the evidence did or did not support it.

If the full hypothesis cannot be supported, the minimum fallback study is a validated open-source comparison of data-only and Galerkin-informed POD models for forward and inverse idealized stenosis flow. If inverse identification is not reliable, the project may narrow to forward field reconstruction and measurement-design analysis. If a global basis fails, the focus may shift to local bases or to documenting the geometry-induced rank limitation.

7. Development and Implementation Plan

This section defines technical gates rather than a fixed calendar. The engineering notebook will record dates, commits, computational cost, and deviations from the plan.

Gate	Objective	Required evidence	Decision enabled
0. Data and literature audit	Confirm task overlap, public data, licenses, and feasible first geometry.	Dataset manifests, reference matrix, and selected benchmark.	Reuse public data, generate CFD, or adopt a hybrid.
1. Full-order benchmark	Verify idealized stenosis solver and post-processing.	Mesh study, pressure/WSS/separation comparison, conservation checks.	Accept full-order source for snapshots.
2. Snapshot pipeline	Automate geometry, simulation, extraction, and metadata.	Repeatable sweep with hashes, units, and failure logging.	Scale the offline dataset.
3. POD and data-only baselines	Measure low-rank structure and simple surrogate performance.	Singular values, reconstruction errors, POD-RBF and POD-NN results.	Use global POD, local POD, or revise geometry mapping.
4. PG-PINN forward model	Implement reduced residual training.	Coefficient and field comparison against held-out cases.	Determine whether reduced physics adds value.
5. Inverse severity model	Recover η from controlled limited observations.	Noise, sensor-ablation, identifiability, and uncertainty results.	Expand to length or location.
6. Residual closure	Correct a clearly defined model discrepancy.	Ablation, stability, held-out performance, correction magnitude.	Retain, constrain further, or reject closure.
7. Demonstration and release	Package verified workflow and interface.	Reproducible examples, documentation, license review, archived release.	Public repository and technical communication.

Table 5. Stage-gated implementation plan. No gate is passed solely because code runs.

7.1 Software organization

The repository should separate geometry generation, CFD interfaces, snapshot processing, POD, reduced operators, neural models, inverse experiments, closure models, validation, and visualization. Configuration files should define each experiment. Large fields should remain outside Git history and be represented by manifests containing source, checksum, units, geometry parameters, solver settings, and creation date. Every figure intended for a report should be reproducible from a script and data manifest.

7.2 Engineering notebook relationship

This project definition is a controlled overview. It should not be expanded with every temporary idea. The engineering notebook will record dated hypotheses, failed runs, code-generation assistance, parameter choices, unresolved questions, and reasons for changes. When a notebook decision changes the project scope, this document should receive a new version rather than being silently edited.

8. Risks and Mitigation

Risk	Consequence	Early indicator	Mitigation / narrowing action
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Risk	Consequence	Early indicator	Mitigation / narrowing action
No suitable public volumetric dataset	Insufficient snapshots for POD-Galerkin training.	Public data contain only images, centerline metrics, or inaccessible fields.	Generate a small verified idealized CFD set; use public data for separate geometry or validation tasks.
Geometry mapping dominates error	Reduced basis mixes interpolation error with flow variability.	Mapped full-order fields disagree before POD.	Use common topology, improve registration, restrict geometry family, or report mapping error explicitly.
Global basis is not compact	Online model loses speed or accuracy.	Slow singular-value decay and severity-dependent reconstruction failure.	Use local/clustered POD or narrow the parameter domain.
Inverse problem is non-identifiable	Different lesions yield similar observations.	Multimodal posteriors or unstable estimates under small noise.	Add measurements, infer distributions, or redefine the task as measurement-design analysis.
Physics loss adds no value	PG-PINN is more complex than a data-only surrogate without benefit.	Equivalent or worse held-out error and data efficiency.	Retain simpler model and document the comparison.
Closure replaces rather than corrects physics	Good fit with poor stability or generalization.	Large correction magnitude or residual cancellation.	Constrain the closure, isolate discrepancy sources, or remove it.
PINN optimization is unstable	Training is sensitive to loss weights or initialization.	High variance across seeds and inconsistent residual convergence.	Normalize variables, stage training, use adaptive weights, and preserve non-PINN baselines.
Near-wall quantities are inaccurate	WSS conclusions are unreliable despite low global error.	Large gradient error at the throat or wall.	Increase basis resolution near walls, use weighted norms, or limit claims to velocity/pressure.
Compute or storage exceeds resources	Offline data generation stalls.	Single-case runtime or storage prevents parameter coverage.	Use wedge/axisymmetric models, smaller controlled campaigns, public subsets, or institutional compute later.
Unverified AI-assisted output enters the code or report	False citations, incorrect derivations, or hidden software defects.	Code passes superficial tests but fails benchmarks; sources cannot be verified.	Require human review, unit tests, benchmark validation, and independent source checking.

Table 6. Initial risk register. Risks will be updated through the engineering notebook and formal revisions.

9. Scope, Exclusions, and Claim Boundaries

9.1 Included in the initial research scope

- A controlled family of idealized or registered arterial stenosis geometries.
- Velocity and pressure as primary reduced fields.
- Stenosis severity as the first inverse target.
- POD, Galerkin-reduced residuals, data-only baselines, and a later constrained closure.
- Sparse, noisy, or missing synthetic observations used for repeatable inverse tests.
- A public-data audit and use of suitable open datasets when licenses and fields permit.
- An open-source research codebase after validation and release review.

9.2 Explicit exclusions

- No clinical diagnosis, treatment recommendation, patient-safety claim, or claim of medical-device status.
- No claim of universal performance on arbitrary patient anatomy.
- No patient data in the initial implementation.
- No assumption that ordinary systemic blood pressure uniquely determines local stenosis hemodynamics.
- No fluid-structure interaction, compliant walls, or full cardiovascular network in the initial model.
- No claim that a ROM is more accurate than the high-fidelity data used to construct it.
- No guaranteed real-time speed, accuracy, novelty, or publication outcome before evidence exists.
- No use of the term digital twin as a technical claim in this project definition.

9.3 Demonstration interface

A frontend is considered a demonstration and validation layer, not the scientific contribution by itself. It should be built after the model has passed held-out tests. The interface may expose input parameters, sensor masks, estimated severity, uncertainty, reconstructed fields, and warnings when a case lies outside the trained domain. A polished public interface becomes appropriate only when the underlying model behavior is documented and reproducible.

10. Research Practice, Independence, and Open-Source Intent

10.1 Independent project record

The work is being developed independently from public information and personal resources. This statement is included to establish provenance and current project status, not to imply that independent work is exempt from later institutional, data-use, publication, or licensing requirements. If faculty guidance, formal collaboration, or institutional resources materially shape the work, future versions should update the contributor and project-status record.

10.2 Open-source intent

The intended software outcome is an open-source research implementation. The repository will remain private during initial feasibility work. Public release should follow a validated prototype, source and dataset license review, removal of restricted or nonredistributable material, and preparation of reproducible examples. The software license is not fixed in version 0.1. A permissive license may be appropriate, but the choice should be made after dependency and dataset obligations are known.

10.3 AI-assisted development and authorship responsibility

Generative AI assistants are used to accelerate routine workflow, including code scaffolding, debugging support, literature organization, document structure, and drafting. The research hypothesis, mathematical formulation, method selection, design decisions, source verification, derivations, code review, verification and validation, interpretation, and final authorship responsibility remain with the researcher. AI-generated citations, factual claims, equations, and code will not be accepted without independent checking.

Current publisher policies commonly require disclosure when generative AI materially contributes to manuscript preparation or research methods, while retaining full human responsibility for accuracy and authorship [15]. A future paper will follow the policy of the selected journal and will document AI-assisted code development or analysis in the methods when required. This project will not attempt to log every minor interaction, but consequential AI-assisted code or methodological choices should be traceable through commits, tests, and notebook entries.

11. Immediate Next Actions

1. Create the repository documentation structure and commit this project definition as version 0.1 after replacing the email placeholder and confirming the repository address.
2. Create a literature and dataset matrix covering geometry-aware ROMs, inverse hemodynamics, stenosis datasets, public CFD fields, 4D-flow MRI, and learned ROM closures.
3. Inspect the open data and source code associated with Ye et al. and record exact files, licenses, mesh correspondence, variables, and parameter coverage.
4. Download and inspect the aortic-stenosis centerline dataset and ARCADE metadata without integrating either into the codebase until their roles are defined.
5. Select the first full-order benchmark and solver configuration. If public volumetric data are insufficient, implement an axisymmetric or wedge stenosis case and verify it against published trends.

6. Define the first experiment configuration: severity-only inversion, known Reynolds number, fixed stenosis length, velocity and pressure snapshots, and a small set of synthetic sensors.
7. Implement the POD reconstruction baseline before any neural network.
8. Prepare a concise faculty-facing summary derived from this document rather than sending the full internal record without context.

12. Conclusion

The project will test a specific bridge between projection-based reduced physics and inverse scientific machine learning. The first objective is not a universal clinical model. It is a verified demonstration that geometry-varying stenotic flow can be represented in a common reduced space, constrained by projected equations, and inverted from limited observations at lower online cost than full-order CFD. The second objective is to determine whether a constrained learned residual corrects identifiable model discrepancy without erasing the physical structure that motivated the ROM.

The project is technically plausible because closely related work has demonstrated POD-Galerkin PINNs for inverse Navier-Stokes problems [1], geometry-informed POD for stenosis and bifurcation flow [6], PINN-based cardiovascular parameter recovery from noisy measurements [7–10], and fast data-driven hemodynamic prediction on patient-derived geometry [11]. The exact integration remains unproven. That gap defines the research rather than guaranteeing its outcome.

Results will be judged against simpler models, held-out full-order data, measurement-ablation tests, and transparent computational accounting. If the proposed architecture proves unnecessary or infeasible, the project will preserve the evidence and narrow to the strongest supported contribution. If it succeeds, the result will be a reproducible research method and a demonstration tool that can support later work with more realistic geometry or authorized medical data.

Declaration of Generative AI and AI-Assisted Technologies

During preparation of this project definition, generative AI tools were used for drafting support, organization, source discovery, and document-production assistance. The author reviewed and edited the content, independently checked the cited sources used to support technical claims, and retains responsibility for the project definition. Future research code, analyses, and manuscripts will be subject to independent verification, testing, and the disclosure requirements of the applicable venue.

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Appendix A. Symbols and Parameter Definitions

Symbol	Definition	Notes
η	Area-based stenosis severity	Initial primary inverse parameter.
R, R_t	Reference vessel radius and throat radius	Used in the initial idealized parameterization.
L_s/D	Nondimensional stenosis length	Secondary geometric parameter.
Re	Reynolds number	May be specified through flow rate, diameter, and viscosity.
μ	Parameter vector; also commonly dynamic viscosity in local equations	Context must distinguish parameter-vector and viscosity notation.
u, p	Velocity and pressure fields	Primary reconstructed fields.
ϕ_i, χ_j	Velocity and pressure POD modes	Spatial basis functions.
a, b	Reduced velocity and pressure coefficients	Network outputs or reduced states.

Symbol	Definition	Notes
M	Reduced mass matrix	May depend on geometry and basis treatment.
F_PG	Galerkin-projected reduced dynamics	Contains reduced physical operators.
C ψ	Learned residual closure	Added only after the unclosed model is verified.
y_obs	Available measurement vector	Can contain pressure, velocity, geometry, or derived observations.
m	Observation mask	Indicates which measurements are present.
WSS / OSI	Wall shear stress / oscillatory shear index	Derived quantities; OSI requires time-resolved flow.

Appendix B. Validation Metric Definitions

$$\varepsilon_u = ||u_{\text{FOM}} - u_{\text{ROM}}||_2 / ||u_{\text{FOM}}||_2, \quad (\text{B1})$$

$$\varepsilon_p = ||(p_{\text{FOM}} - \bar{p}_{\text{FOM}}) - (p_{\text{ROM}} - \bar{p}_{\text{ROM}})||_2 / ||p_{\text{FOM}} - \bar{p}_{\text{FOM}}||_2, \quad (\text{B2})$$

$$e_\eta = |\eta_{\text{true}} - \eta_{\text{pred}}|, \quad (\text{B3})$$

$$e_{\Delta p} = |\Delta p_{\text{true}} - \Delta p_{\text{pred}}| / \max(|\Delta p_{\text{true}}|, \varepsilon), \quad (\text{B4})$$

Pressure errors should account for the arbitrary reference pressure where applicable. WSS evaluation should include peak error, surface-mean error, and a spatial distribution metric. Speedup should be reported as the full-order wall time divided by total online time, with preprocessing and reconstruction reported separately. Uncertainty evaluation should report empirical interval coverage and interval width or an equivalent calibration metric.

Appendix C. Initial Decision Record

Decision	Current selection	Revisit condition
Project status	Independent research project separate from the aerospace thesis.	Formal collaboration or thesis status changes.
Initial application	General arterial stenosis using a controlled idealized geometry family.	Data audit identifies a clearly better specific vessel or application.
Primary fields	Velocity and pressure.	Evidence shows another field must be modeled directly.
First inverse target	Stenosis severity.	Severity is not identifiable from the selected observations.
Data source	Public CFD if suitable; otherwise verified local idealized CFD; likely hybrid.	Dataset inspection confirms a complete alternative.
Closure timing	After unclosed PG-PINN baseline.	No earlier exception.
Public communication	Private repository until a verified prototype and release review.	A validated milestone is reached.

Appendix D. Documentation Hierarchy

- Project definition document: stable statement of hypothesis, scope, intended method, claim boundaries, and research gates.
- Engineering notebook: dated record of ideas, source evaluations, experiments, failed attempts, decisions, AI-assisted development concerns, and next steps.

- Software documentation: installation, architecture, configuration, data manifests, API behavior, and reproducible examples.
- Technical report or paper: final method, verified data, experiments, results, limitations, and interpretation after the research is complete.